

Static Stress and Deflection Analysis of a Teakwood Closet Rod

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Executive Summary

This report evaluates the structural behavior of a teakwood closet rod subjected to a static clothing load. The rod was modeled as a solid circular beam supported at both ends by open-top wooden bearing inserts. The objective of the analysis was to estimate bending stress, vertical deflection, and overall structural adequacy using both analytical beam theory and finite element analysis (FEA).

The closet rod has a diameter of 1.25 in and an effective span of 60.5 in after accounting for support contact depth. The applied clothing load was estimated to be 71.75 lbf and modeled as a uniformly distributed load. Analytical calculations predicted a maximum bending stress of approximately 2.830 ksi and a factor of safety of approximately 4.98 relative to the modulus of rupture of teakwood. However, the predicted maximum deflection of 0.969 in exceeds a typical serviceability limit of $L/240$, or 0.252 in, indicating deflection as the governing concern.

A SolidWorks Simulation model was developed to refine the analysis by incorporating geometry, contact conditions, and support interactions. The final SolidWorks result predicted a maximum vertical displacement of 1.108 in and a representative midspan axial stress of 2.847 ksi. The FEA stress result agrees closely with the analytical stress estimate, while the larger deflection reinforces the importance of support conditions and long-term deformation behavior.

Overall, the analysis demonstrates that the closet rod is unlikely to fail in bending under the estimated load but is susceptible to excessive deflection. Consequently, stiffness and long-term deformation are more critical design considerations than strength.

1. Introduction

Closet rods are simple structural elements, yet their behavior can be effectively analyzed using fundamental mechanics of materials principles. When supporting clothing, a rod behaves as a beam subjected to distributed loading, where excessive deflection or insufficient support can lead to sag, permanent deformation, support damage, or eventual failure.

The purpose of this project is to evaluate the structural performance of a household closet rod using a simplified engineering approach. The analysis integrates analytical beam calculations, Python-based computation, CAD modeling, and finite element modeling in SolidWorks. Rather than predicting exact failure conditions, the goal is to estimate stress, deflection, and safety margins under a realistic static loading scenario.

This combined approach allows for both theoretical validation and practical insight into how geometry, material properties, and support conditions influence structural behavior.

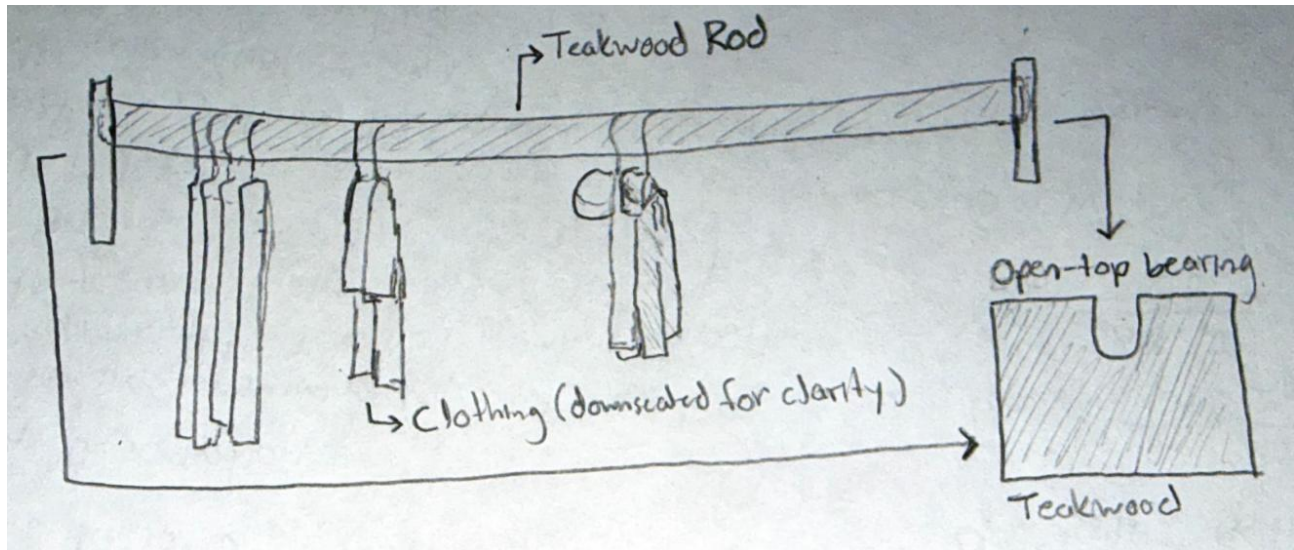


Figure 1. Concept sketch of the closet rod system showing the physical rod, open-top wooden bearing insert, and clothing load.

2. System Description

The system analyzed consists of a solid circular teakwood closet rod supported at both ends by open-top wooden bearing inserts. The rod rests within the supports rather than being rigidly fixed, resulting in partial restraint and uncertainty in rotational boundary conditions. To account for this uncertainty, the rod was conservatively idealized as a simply supported beam for the initial analytical evaluation.

The total rod length is approximately 62 in. However, a portion of the rod is embedded within each support, reducing the effective unsupported span. Accounting for an estimated bearing contact depth of 0.75 in per side, the effective span is calculated using Equation (1).

$$L_e = 62 - 2(0.75) = 60.5 \text{ in} \quad (1)$$

The primary geometric parameters are summarized in Table 1.

Table 1. Rod geometry.

Quantity	Symbol	Value
Rod material	—	Teakwood
Total rod length	L	62 in
Effective bearing contact per side	—	0.75 in
Effective span	L_e	60.5 in
Rod diameter	d	1.25 in

The applied loading was estimated based on the weight of clothing supported by the rod. Individual clothing items were grouped, numbered, and assigned representative weights using averages in Appendix A. This resulted in the total estimated load being shown in Equation (2).

$$W = 71.75 \text{ lbf} \quad (2)$$

For analytical simplicity, this load was modeled as static and uniformly distributed across the effective span. While this assumption neglects localized hanger forces and uneven distribution, it provides a reasonable first-order approximation of global bending behavior.

The distributed load was calculated using Equation (3), resulting in the value shown in Equation (4).

$$w = \frac{W}{L_e} \quad (3)$$

$$w = \frac{71.75}{60.5} = 1.186 \text{ lbf/in} \quad (4)$$

The loading assumptions used throughout the analysis are summarized in Table 2. This model is applied consistently in both the analytical calculations and the numerical simulation to ensure comparability between methods.

Table 2. Loading assumptions.

Quantity	Symbol	Value
Total estimated clothing load	W	71.75 lbf
Effective span	L_e	60.5 in
Distributed load	w	1.186 lbf/in
Load type	—	Static, uniformly distributed
Dynamic effects	—	Neglected

3. Material Properties and Assumptions

Teakwood was modeled using representative published material properties taken from The Wood Database [1]. The main values used in the analysis were the elastic modulus, modulus of rupture, and an assumed Poisson's ratio. These values are summarized in Table 3.

Table 3. Simplified teakwood material properties.

Property	Symbol	Value
Elastic modulus	E	1.781×10^6 psi
Modulus of rupture	S_{mor}	14,080 psi
Poisson's ratio	ν	0.325, assumed

The elastic modulus was used to calculate beam deflection, while the modulus of rupture was used as an approximate bending failure stress. The modulus of rupture is not equivalent to ductile yield strength. Wood does not yield in the same way as steel, so the factor of safety was calculated relative to bending rupture rather than yielding [2].

Several simplifying assumptions were made:

- The rod was modeled as a straight, solid, circular beam.
- The material was treated as homogeneous and isotropic.
- The clothing load was treated as static and uniformly distributed.
- Dynamic effects from pulling clothes or impact loading were neglected.
- The support inserts were idealized as end supports with 0.75 in effective contact length per side.
- Long-term creep was not directly simulated in SolidWorks.
- The FEA material model was used for approximate stiffness and stress visualization, not exact wood failure prediction.

Real wood is anisotropic, grain-dependent, moisture-sensitive, and naturally variable. Therefore, the results should be interpreted as a simplified engineering estimate rather than an exact prediction of real-world failure.

4. Analytical Beam Model

The initial analysis models the rod as a simply supported beam subjected to a uniformly distributed load. This assumption provides a conservative estimate of structural response by allowing free rotation at the supports, which maximizes deflection relative to more restrained boundary conditions.

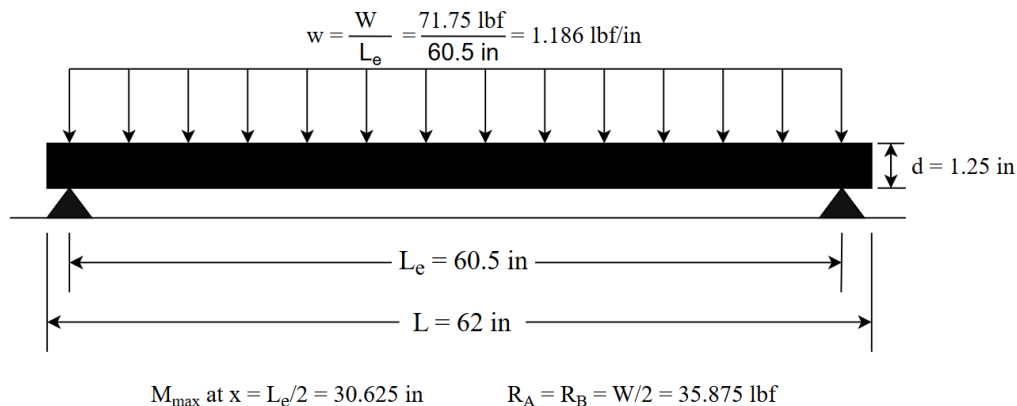


Figure 2. Simplified simply supported beam model used for preliminary hand calculations.

Using standard beam theory, expressions for support reactions, bending moment, stress, and deflection were derived. These results form the baseline for comparison with finite element results and provide insight into expected structural behavior under the applied loading conditions.

For a simply supported beam with total load W uniformly distributed over span L_e , the reaction forces at each support are equal due to symmetry.

$$R_A = R_B = \frac{W}{2} \quad (5)$$

$$R_A = R_B = \frac{71.75}{2} = 35.875 \text{ lbf} \quad (6)$$

The maximum bending moment for a uniformly distributed load occurs at the midpoint of the beam. Since $w = W/L_e$, the maximum moment can be expressed in terms of either w or W [3].

$$x = \frac{L_e}{2} = 30.25 \text{ in} \quad (7)$$

$$M_{max} = \frac{wL_e^2}{8} = \frac{WL_e}{8} \quad (8)$$

$$M_{max} = \frac{(71.75)(60.5)}{8} = 542.6 \text{ lbf} \cdot \text{in} \quad (9)$$

For a solid circular cross-section, the area moment of inertia and outer-fiber distance are given by Equations (10) and (12).

$$I = \frac{\pi d^4}{64} \quad (10)$$

$$I = \frac{\pi(1.25)^4}{64} = 0.1198 \text{ in}^4 \quad (11)$$

$$c = \frac{d}{2} = 0.625 \text{ in} \quad (12)$$

The maximum bending stress occurs at the outer fiber at the location of maximum moment.

$$\sigma_{max} = \frac{M_{max}c}{I} \quad (13)$$

$$\sigma_{max} = \frac{(542.6)(0.625)}{0.1198} = 2830 \text{ psi} \quad (14)$$

$$\sigma_{max} = 2.83 \text{ ksi} \quad (15)$$

Using the teakwood modulus of rupture, $S_{mor} = 14,080 \text{ psi}$, the factor of safety relative to bending rupture is calculated as follows.

$$FS = \frac{S_{MOR}}{\sigma_{max}} \quad (16)$$

$$FS = \frac{14080}{2830} = 4.98 \quad (17)$$

For a simply supported beam with a uniformly distributed load, the maximum vertical deflection occurs at midspan [3].

$$\delta_{max} = \frac{5wL_e^4}{384EI} \quad (18)$$

$$\delta_{max} = \frac{5(1.186)(60.5)^4}{384(1.781 \times 10^6)(0.1198)} \quad (19)$$

$$\delta_{max} = 0.969 \text{ in} \quad (20)$$

A common serviceability-style deflection limit is $L/240$ [4]. For the effective span used in this analysis, the allowable deflection is calculated in Equation (21).

$$\delta_{allow} = \frac{L_e}{240} = \frac{60.5}{240} = 0.252 \text{ in} \quad (21)$$

$$\frac{\delta_{max}}{\delta_{allow}} = \frac{0.969}{0.252} = 3.84 \quad (22)$$

The predicted deflection exceeds the selected serviceability limit by nearly a factor of four, indicating that deflection is likely the governing design constraint rather than material failure.

The estimated bending rupture load was calculated by rearranging the bending stress relation using the modulus of rupture as the limiting stress.

$$W_{rupture} = \frac{8S_{MOR}I}{cL_e} \quad (23)$$

$$W_{rupture} = \frac{8(14080)(0.1198)}{(0.625)(60.5)} = 357 \text{ lbf} \quad (24)$$

The load corresponding to the $L/240$ deflection limit was estimated by scaling the applied load by the ratio of allowable to predicted deflection.

$$W_{L/240} = W\left(\frac{\delta_{allow}}{\delta_{max}}\right) \quad (25)$$

$$W_{L/240} = 71.75\left(\frac{0.252}{0.969}\right) = 18.7 \text{ lbf} \quad (26)$$

A simplified long-term creep estimate was also calculated using a multiplier of 1.5. This is not a true time-dependent creep model but provides a conservative indication of sustained-load deformation [5].

$$\delta_{long-term} = 1.5\delta_{max} = 1.5(0.969) = 1.45 \text{ in} \quad (27)$$

5. Python Calculation Tool

A Python script was created to automate the analytical beam calculations and compare them with the SolidWorks Simulation results. The script takes the rod geometry, material properties, estimated clothing load, serviceability limit, and FEA comparison values as inputs. It then calculates section properties, support reactions, maximum bending moment, bending stress, factor of safety, maximum vertical deflection, load capacity estimates, and a center-support comparison.

```
SIMPLY SUPPORTED BEAM RESULTS
-----
Reaction at each support:          35.875 lbf
Max moment location:              30.250 in
Maximum bending moment:           542.61 lbf-in
Maximum bending stress:           2.830 ksi
Factor of safety vs MOR:          4.98
Maximum vertical deflection:       0.969 in
L/240 deflection limit:            0.252 in
Deflection / allowable:           3.85
Long-term deflection estimate:     1.454 in

LOAD CAPACITY ESTIMATES
-----
Estimated bending rupture load:    357.0 lbf
Load at L/240 deflection:          18.7 lbf

SOLIDWORKS FEA COMPARISON
-----
FEA vertical deflection:           1.108 in
FEA midspan axial stress:          2.847 ksi
FEA factor of safety vs MOR:       4.95
FEA reaction at A:                 35.875 lbf
FEA reaction at B:                 35.875 lbf
FEA total vertical reaction:       71.750 lbf
Reaction error vs load:            0.000 lbf

INTERPRETATION
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Strength: bending rupture is not predicted under the estimated load.
Serviceability: deflection exceeds the selected L/240 limit.
Overall: the rod is strength-acceptable, but deflection/serviceability controls.
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Figure 3. Python output for the teakwood closet rod preliminary beam calculation.

The Python tool was useful because it made the project parametric. If the rod diameter, effective span, or total clothing load is changed, the stress and deflection values can be recalculated quickly. This is especially useful for comparing design changes such as increasing rod diameter, reducing span length, or adding a center support.

6. SolidWorks CAD and Simulation Model

A simplified SolidWorks assembly was created to represent the rod and the open-top wooden bearing inserts. The model included three primary components: the teakwood rod, the left bearing insert, and the right bearing insert. The rod was modeled as a solid 1.25 in diameter cylinder, and the supports were modeled as open-top wooden cradle-style bushings.

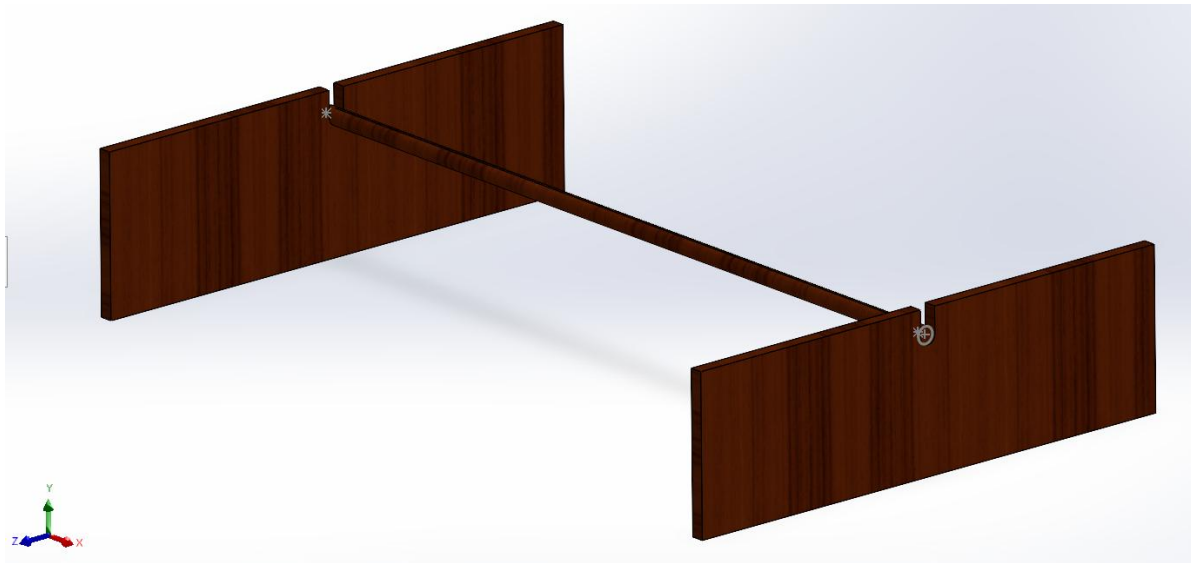


Figure 4. SolidWorks CAD model of the closet rod and open-top wooden support inserts.

The SolidWorks Simulation study was set up as a static structural analysis. The bushing back faces were fixed to represent the closet structure supporting the inserts. The rod-to-bushing interfaces were modeled using contact interactions so that the rod could bear against the lower cradle surfaces without being bonded to the supports.

The total clothing load of 71.75 lbf was applied downward to the rod as a distributed load. A simplified isotropic teakwood material was assigned to the rod, using the elastic modulus and density values listed earlier. The model was meshed with local refinement near the rod-bushing contact regions.

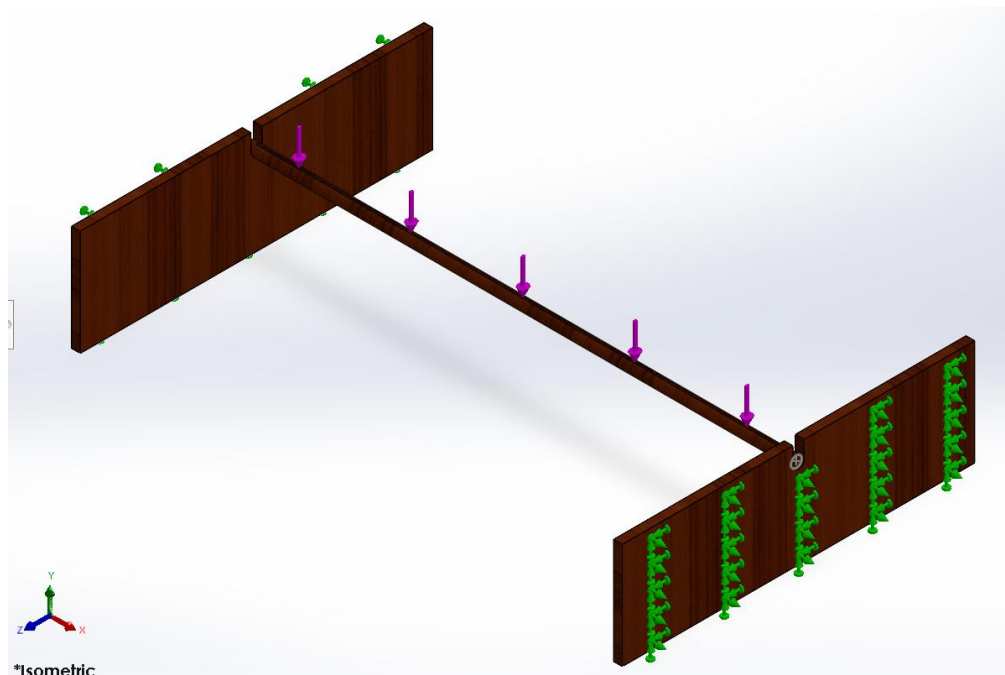


Figure 5. SolidWorks Simulation setup showing applied clothing load, fixed bushing faces, and contact regions.

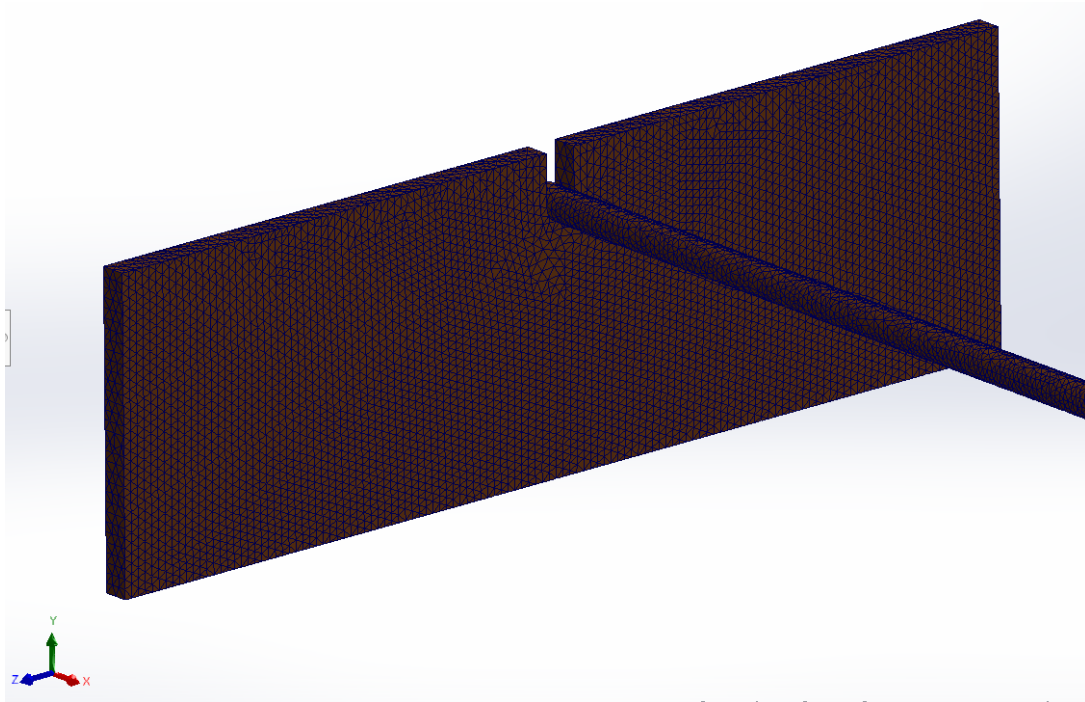


Figure 6. Mesh used for the SolidWorks Simulation study, with refinement near the rod-bushing contact regions. Shows only one side for mesh clarity.

7. FEA Results

Finite element analysis was conducted using SolidWorks Simulation to better capture the effects of geometry, contact behavior, and support conditions that are not fully represented in the simplified analytical model. The primary quantity of interest was the maximum vertical displacement of the rod under the applied load, since preliminary calculations indicated that deflection was more critical than bending rupture.

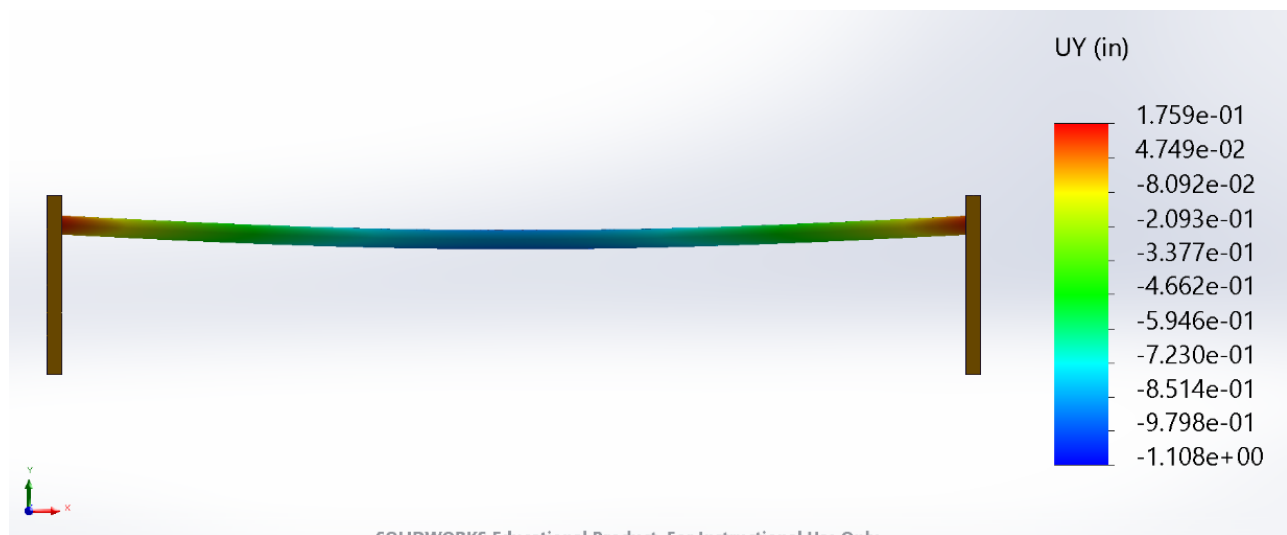


Figure 7. Front view of vertical displacement plot of the rod under distributed clothing load.

The FEA model predicted the maximum vertical displacement shown in Equation (28).

$$\delta_{FEA} = 1.108 \text{ in} \quad (28)$$

This value is slightly larger than the analytical prediction of 0.969 in. The FEA displacement was slightly larger than the analytical prediction of 0.969 in. This difference is likely due to modeling details such as contact behavior, effective support location, load application, and local support geometry. Because the model includes open-bushing contact rather than idealized point supports, the FEA result was treated as a secondary comparison rather than an exact physical prediction.

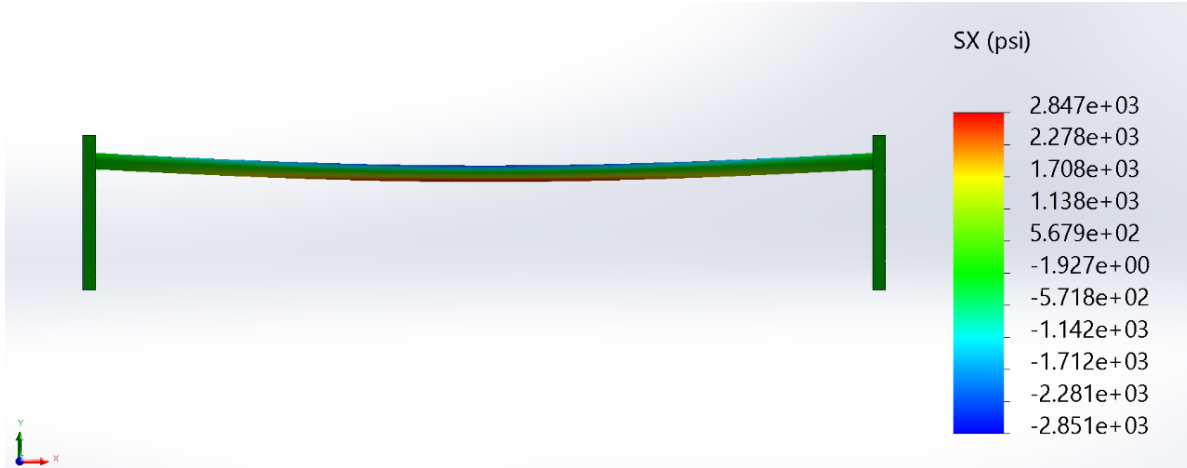


Figure 8. Axial stress plot from the SolidWorks Simulation study. Local contact stresses near the supports were not used as the primary bending-stress comparison.

The representative midspan axial stress obtained from the simulation is shown in Equation (29).

$$\sigma_{FEA} = 2.847 \text{ ksi} \quad (29)$$

This value closely matches the analytical prediction of 2.830 ksi, indicating strong agreement between the beam theory and the numerical model for stress estimation. It is important to note that localized stress concentrations near the supports were not used for comparison, as these are influenced by contact geometry and do not represent global bending behavior.

Reaction forces at the supports were evaluated to verify equilibrium. For a symmetric load, the expected reaction at each support is shown in Equation (30).

$$R = \frac{W}{2} = 35.875 \text{ lbf} \quad (30)$$

Component	Selection	Component	Selection
Sum X:	0.0031979	Sum X:	-0.0031636
Sum Y:	35.875	Sum Y:	35.875
Sum Z:	0.047567	Sum Z:	-0.047582
Resultant:	35.875	Resultant:	35.875

Figures 9 and 10. Reaction force outputs at points A and B, respectively, used to verify that the support reactions balance the applied clothing load.

The FEA results showed reaction forces consistent with this value, confirming that the applied loading and boundary conditions were correctly defined. The total vertical reaction is approximately equal to the applied 71.75 lbf clothing load, neglecting rod self-weight unless gravity is included separately.

The close agreement in stress results confirms that classical beam theory provides an accurate estimate of bending stress for this system. However, the increased deflection observed in the FEA model highlights the sensitivity of deflection to support conditions and boundary assumptions. This reinforces that deflection, rather than stress, governs the structural performance of the closet rod under the given loading conditions.

8. Results Comparisons

The primary analytical and simulation results are summarized in Table 4.

Table 4. Analytical and FEA result comparison.

Quantity	Simply Supported	SolidWorks FEA
Effective span	60.5 in	60.5 in
Total load	71.75 lbf	71.75 lbf
Reaction per support	35.875 lbf	35.875 lbf
Maximum bending stress	2.830 ksi	2.847 ksi
Maximum vertical deflection	0.969 in	1.108 in
Factor of safety vs MOR	4.98	4.95

The analytical and FEA stress values agree closely, which supports the use of the simplified beam model for estimating bending stress. The deflection results differ more noticeably, which is expected because displacement is more sensitive to support stiffness, contact behavior, and idealization of boundary conditions.

The simply supported beam model provides a useful analytical baseline because it assumes free rotation at the supports and can be evaluated with standard closed-form beam equations. The actual open-bushing support condition is more complex because it depends on contact geometry, bearing depth, friction, and fit between the rod and the wooden inserts. Therefore, the SolidWorks contact model was used as a secondary comparison rather than as an exact physical prediction. Both the analytical and FEA results indicate that the rod has a relatively high factor of safety against bending rupture, while the predicted vertical deflection exceeds the selected $L/240$ serviceability benchmark. This suggests that the rod is more likely to be limited by sag and possible long-term deformation than by immediate bending failure.

9. Discussion

The results indicate that the primary limitation of the closet rod is serviceability rather than strength. The analytical model predicts a factor of safety significantly greater than with respect to bending failure, suggesting that the rod is unlikely to fracture under the estimated loading conditions.

However, the predicted deflection exceeds commonly accepted serviceability limits, indicating that noticeable sag is expected during normal use. This discrepancy highlights the importance of stiffness in practical applications, where user perception and long-term deformation often govern performance.

The finite element model provides additional insight by incorporating support geometry and contact interactions. While the predicted stress values closely match the analytical results, the deflection differs due to the influence of support conditions that are not fully captured in the idealized beam model. The largest source of uncertainty in the analysis is the support condition, which influences deflection more strongly than stress.

Several limitations should be noted. The material was modeled as homogeneous and isotropic, whereas real wood exhibits anisotropic behavior influenced by grain direction, knots, moisture content, age, and natural defects. Additionally, the loading was simplified as uniformly distributed, despite real-world variability from individual hangers. In practice, long-term creep and localized bearing deformation may further increase deflection beyond both analytical and FEA predictions.

Despite these limitations, the combined analytical and numerical approach provides a reliable first-order assessment of performance and identifies deflection as the governing design concern. The results suggest that increasing stiffness would be more important than increasing strength. The most effective design improvement would likely be adding a center support, reducing the unsupported span, or increasing the rod diameter.

10. Recommendations

Based on the analysis, the following recommendations are made:

- Add a center support if noticeable sag is unacceptable.
- Reduce the effective span where possible.
- Use a larger diameter rod if replacing the current rod.

- Avoid large uneven point loads or concentrated heavy clothing sections.
- Treat long-term sag and creep as the main practical concern rather than immediate fracture.
- If a more realistic failure model is desired, evaluate local bearing stresses, support loosening, and wood creep behavior.

A center support would be especially effective because beam deflection scales strongly with span length. For a uniformly loaded beam, deflection is proportional to L^4 . Therefore, reducing the span by half can reduce deflection by roughly a factor of 16.

11. Conclusion

This study evaluated the structural performance of a teakwood closet rod under a realistic clothing load using analytical and numerical methods. The rod has a 1.25 in diameter and a 60.5 in effective span, and it was analyzed under an estimated 71.75 lbf uniformly distributed load.

The analytical calculations predicted a maximum bending stress of approximately 2.83 ksi and a factor of safety of approximately 4.98 relative to teakwood's modulus of rupture. This indicates that immediate bending rupture is unlikely under the estimated loading conditions. However, the predicted simply supported deflection was approximately 0.969 in, which exceeds the selected $L/240$ serviceability limit of 0.252 in.

The finite element results confirm this conclusion and demonstrate that support conditions and contact behavior significantly influence overall deformation. The close agreement between analytical and FEA stress results validates the modeling assumptions, while differences in deflection highlight the limitations of simplified boundary conditions.

Overall, the analysis shows that the closet rod is strength-acceptable under the estimated static load, but potentially serviceability-limited due to deflection. The most effective improvement would be to reduce the span by adding a center support, or to increase stiffness by using a larger-diameter rod.

These findings emphasize the importance of considering both strength and serviceability in evaluating seemingly simple structural systems.

References

- [1] The Wood Database. “Teak.” Accessed May 2026. <https://www.wood-database.com/teak/>
- [2] The Wood Database. “Modulus of Rupture.” Accessed May 2026. <https://www.wood-database.com/wood-articles/modulus-of-rupture/>
- [3] MechaniCalc. “Beam Stress & Deflection.” Accessed May 2026. <https://mechanicalc.com/reference/beam-analysis>
- [4] American Wood Council. “Span Table Tutorial.” Accessed May 2026. <https://awc.org/resource-hub/span-table-tutorial/>
- [5] American Wood Council. “What Is Creep and How Can I Address It?” Accessed May 2026. <https://awc.org/faq/what-is-creep-and-how-can-i-address-it/>

AI Statement

ChatGPT was used to help maintain project workflow, check calculations, troubleshoot SolidWorks and Python code, and improve report organization. Microsoft 365 Copilot was used to help with report formatting, writing clarity, and document polish in Microsoft Word. Final assumptions, calculations, CAD/FEA setup, numerical results, interpretation, and report edits were generated and controlled by the author.

Appendix A: Preliminary Calculations

Given: $E = 1.781 \times 10^6 \text{ psi}$, $S_{\text{MOR}} = 14080 \text{ psi}$, $W = 71.25 \text{ lbf}$, $L_c = 60.5 \text{ in}$, $d = 1.25 \text{ in}$

$w = \frac{W}{L_c} = \frac{71.25 \text{ lbf}}{60.5 \text{ in}} = 1.186 \text{ lbf/in}$ (Distributed load) $R_A = R_B = \frac{W}{2} = \frac{71.25 \text{ lbf}}{2} = 35.875 \text{ lbf}$ (Reaction Force at each end)

$I = \frac{\pi d^4}{64} = \frac{\pi (1.25 \text{ in})^4}{64} = 0.1198 \text{ in}^4$ $M_{\text{max}} = \frac{wL_c^2}{8} = \frac{WL_c}{8} = \frac{(71.25 \text{ lbf})(60.5 \text{ in})}{8} = 542.6 \text{ lbf}\cdot\text{in}$ (Maximum moment at midspan)

$\sigma_{\text{max}} = \frac{M_{\text{max}} c}{I} = \frac{M_{\text{max}} d}{2I} = \frac{(542.6 \text{ lbf}\cdot\text{in})(1.25 \text{ in})}{2(0.1198 \text{ in}^4)} = 2830 \text{ psi} = 2.830 \text{ ksi}$; Maximum Bending Stress

$FS_{\text{MOR}} = \eta_{\text{MOR}} = \frac{S_{\text{MOR}}}{\sigma_{\text{max}}} = \frac{14080 \text{ psi}}{2830 \text{ psi}} = 4.98$; Factor of safety between modulus of rupture, which can be used for wood bending failure, and the maximum experienced bending stress

$\delta_{\text{max}} = \frac{5wL_c^4}{384EI} = \frac{5(1.186 \text{ lbf/in})(60.5 \text{ in})^4}{384(1.781 \times 10^6 \text{ psi})(0.1198 \text{ in}^4)} = 0.969 \text{ in}$; maximum deflection of rod under experienced loading

$\delta_{\text{allow}} = \frac{L_c}{240} = \frac{60.5 \text{ in}}{240} = 0.252 \text{ in}$ $\frac{\delta_{\text{max}}}{\delta_{\text{allow}}} = \frac{0.969 \text{ in}}{0.252 \text{ in}} = 3.84$, $\delta_{\text{max}} > \delta_{\text{allow}}$; Sag is more critical than failure using $\frac{L}{240}$ serviceability values

$W_{\text{rupture}} = \frac{8S_{\text{MOR}}I}{cL_c} = \frac{8(14080 \text{ psi})(0.1198 \text{ in}^4)}{(0.625 \text{ in})(60.5 \text{ in})} = 357 \text{ lbf}$; Load required to rupture rod

$W_{L/240} = W\left(\frac{\delta_{\text{allow}}}{\delta_{\text{max}}}\right) = 71.25 \text{ lbf}\left(\frac{0.252 \text{ in}}{0.969 \text{ in}}\right) = 18.7 \text{ lbf}$; Load required to push rod past serviceability limit and into sag criterion

$\delta_{\text{creep}} \approx 1.5 \delta_{\text{max}} = 1.5(0.969 \text{ in}) = 1.45 \text{ in}$; simplified long-term deflection estimate

Key Values:

$w = 1.186 \text{ lbf/in}$	$\sigma_{\text{max}} = 2.830 \text{ ksi}$	$\delta_{\text{max}} = 0.969 \text{ in}$
$R_A = R_B = 35.875 \text{ lbf}$	$FS_{\text{MOR}} = 4.98$	$\delta_{\text{allow}} = 0.252 \text{ in}$
$M_{\text{max}} = 542.6 \text{ lbf}\cdot\text{in}$		

Sources / reasonings:

- S_{MOR} , modulus of rupture, is described by the Wood Database as an alternative ultimate/yield strength for wood because it behaves differently than steel in bending.
- Area moment of inertia, I , is for a solid cylinder in one-directional bending.
- Maximum moment and deflection, M_{max} and δ_{max} , are previously derived equations for simply supported beams experiencing a uniform distributed load.
- Allowable estimates ($\frac{L}{240}$ serviceability) are experimentally derived from the American Wood Council.

Appendix Figure A1. Preliminary hand calculations for the closet rod beam model, including maximum moment, bending stress, deflection, factor of safety, rupture load, and serviceability deflection load.

Appendix Table A1. Clothing load estimates, using rough averages from scale.

Clothing type	Count	Assumed Weight	Subtotal
Shirts	40	0.375 lb	15 lb
Pants	15	1.25 lb	18.75 lb
Hoodies	20	1.5 lb	30 lb
Hangers	80	0.1 lb	8 lb
Total	—	—	71.75 lbf

Appendix B: Python Code

The following Python script was used to automate the analytical calculations. The code mirrors the hand-calculation workflow and was used to verify the values reported in the analytical model and results comparison sections.

```
import math

# 1. INPUTS
# Geometry
L_total = 62.0          # total rod length [in]
bearing_contact = 0.75  # effective support contact per side [in]
d = 1.25                # rod diameter [in]

# Effective span
L_e = L_total - 2 * bearing_contact

# Loading
W = 71.75               # total estimated clothing load [lbf]
w = W / L_e             # uniformly distributed load [lbf/in]

# Teakwood material properties
E = 1.781e6             # elastic modulus [psi]
MOR = 14080             # modulus of rupture [psi]
nu = 0.325              # assumed Poisson's ratio, used in FEA only

# Serviceability / creep assumptions
deflection_limit_ratio = 240
creep_multiplier = 1.5

# SolidWorks FEA results
fea_deflection = 1.108  # max vertical displacement [in]
fea_stress = 2847       # representative midspan axial stress [psi]
fea_reaction_A = 35.875 # vertical reaction at support A [lbf]
fea_reaction_B = 35.875 # vertical reaction at support B [lbf]

# 2. SECTION PROPERTIES
I = math.pi * d**4 / 64 # area moment of inertia [in^4]
c = d / 2               # outer fiber distance [in]

# 3. SIMPLY SUPPORTED BEAM MODEL
R_A = R_B = W / 2       # Support reactions [lbf]
x_Mmax = L_e / 2
M_max = W * L_e / 8      # Maximum bending moment [lbf*in]

# Bending stress and factor of safety
sigma_max = M_max * c / I
FS_MOR = MOR / sigma_max
delta_max = (5 * w * L_e**4) / (384 * E * I) # Maximum vertical deflection [in]

# Serviceability check
delta_allow = L_e / deflection_limit_ratio
delta_ratio = delta_max / delta_allow
delta_creep = creep_multiplier * delta_max # Simplified long-term creep estimate [in]

# Load capacity estimates
W_rupture = (8 * MOR * I) / (c * L_e)
W_L240 = W * (delta_allow / delta_max)

# 4. SOLIDWORKS FEA COMPARISON
fea_FS = MOR / fea_stress
fea_total_reaction = fea_reaction_A + fea_reaction_B
reaction_error = fea_total_reaction - W

# 7. OUTPUT
def section(title):
    print(f"\n{title}")
    print("-" * len(title))
```

```

def line(label, value="", unit="", decimals=3):
    if isinstance(value, float):
        text = f"{value:.{decimals}f}"
    else:
        text = str(value)
    print(f"{label:<35} {text:>12} {unit}")

print("\nCLOSET ROD STRUCTURAL ANALYSIS")
print("=" * 36)

section("INPUTS")
line("Total rod length:", L_total, "in")
line("Effective support contact:", bearing_contact, "in per side")
line("Effective span:", L_e, "in")
line("Rod diameter:", d, "in")
line("Total clothing load:", W, "lbf")
line("Distributed load:", w, "lbf/in")
line("Elastic modulus, E:", f"{E:.3e}", "psi")
line("Modulus of rupture, MOR:", MOR, "psi", decimals=0)

section("SECTION PROPERTIES")
line("Moment of inertia, I:", I, "in^4", decimals=4)

section("SIMPLY SUPPORTED BEAM RESULTS")
line("Reaction at each support:", R_A, "lbf")
line("Max moment location:", x_Mmax, "in")
line("Maximum bending moment:", M_max, "lbf-in", decimals=2)
line("Maximum bending stress:", sigma_max / 1000, "ksi")
line("Factor of safety vs MOR:", FS_MOR, decimals=2)
line("Maximum vertical deflection:", delta_max, "in")
line(f"L/{deflection_limit_ratio} deflection limit:", delta_allow, "in")
line("Deflection / allowable:", delta_ratio, decimals=2)
line("Long-term deflection estimate:", delta_creep, "in")

section("LOAD CAPACITY ESTIMATES")
line("Estimated bending rupture load:", W_rupture, "lbf", decimals=1)
line(f"Load at L/{deflection_limit_ratio} deflection:", W_L240, "lbf", decimals=1)

section("SOLIDWORKS FEA COMPARISON")
line("FEA vertical deflection:", fea_deflection, "in")
line("FEA midspan axial stress:", fea_stress / 1000, "ksi")
line("FEA factor of safety vs MOR:", fea_FS, decimals=2)
line("FEA reaction at A:", fea_reaction_A, "lbf")
line("FEA reaction at B:", fea_reaction_B, "lbf")
line("FEA total vertical reaction:", fea_total_reaction, "lbf")
line("Reaction error vs load:", reaction_error, "lbf")

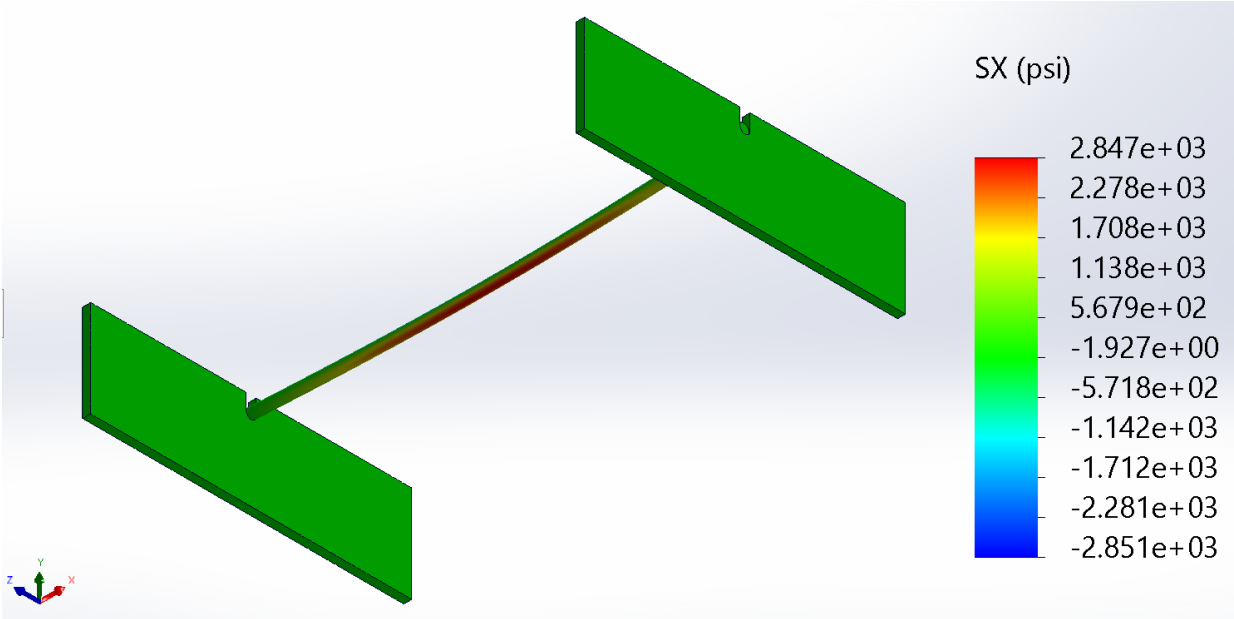
section("INTERPRETATION")
if FS_MOR > 1:
    print("Strength: bending rupture is not predicted under the estimated load.")
else:
    print("Strength: bending rupture is predicted under the estimated load.")

if delta_max > delta_allow:
    print(f"Serviceability: deflection exceeds the selected L/{deflection_limit_ratio} limit.")
else:
    print(f"Serviceability: deflection is below the selected L/{deflection_limit_ratio} limit.")

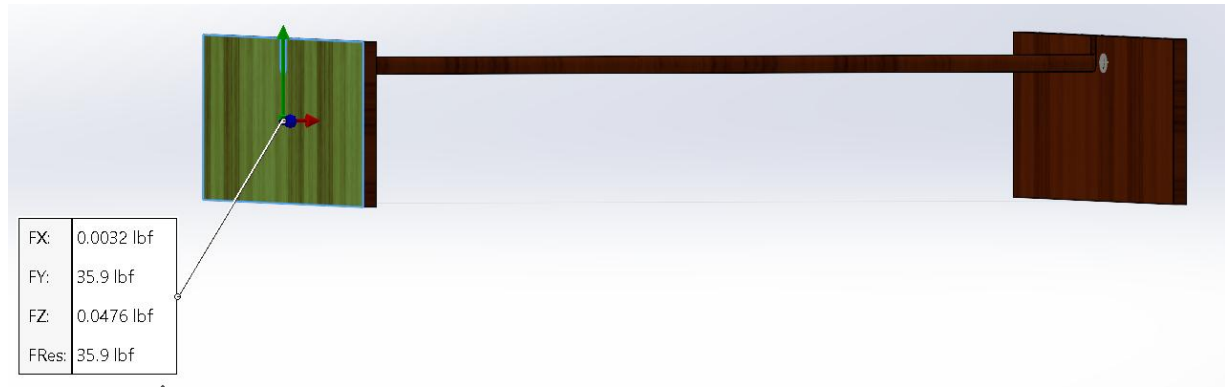
print("Overall: the rod is strength-acceptable, but deflection/serviceability controls.")

```

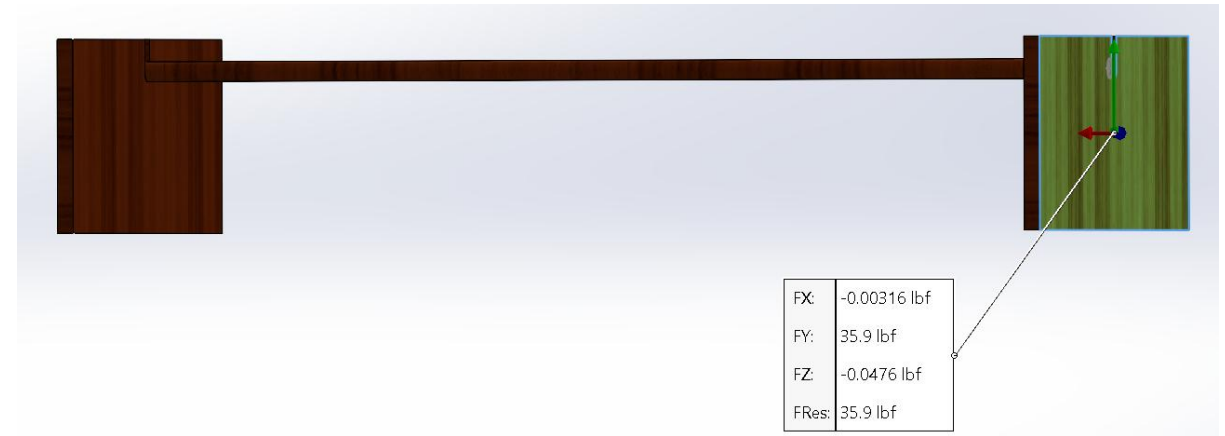
Appendix C: Additional SolidWorks Simulation Figures



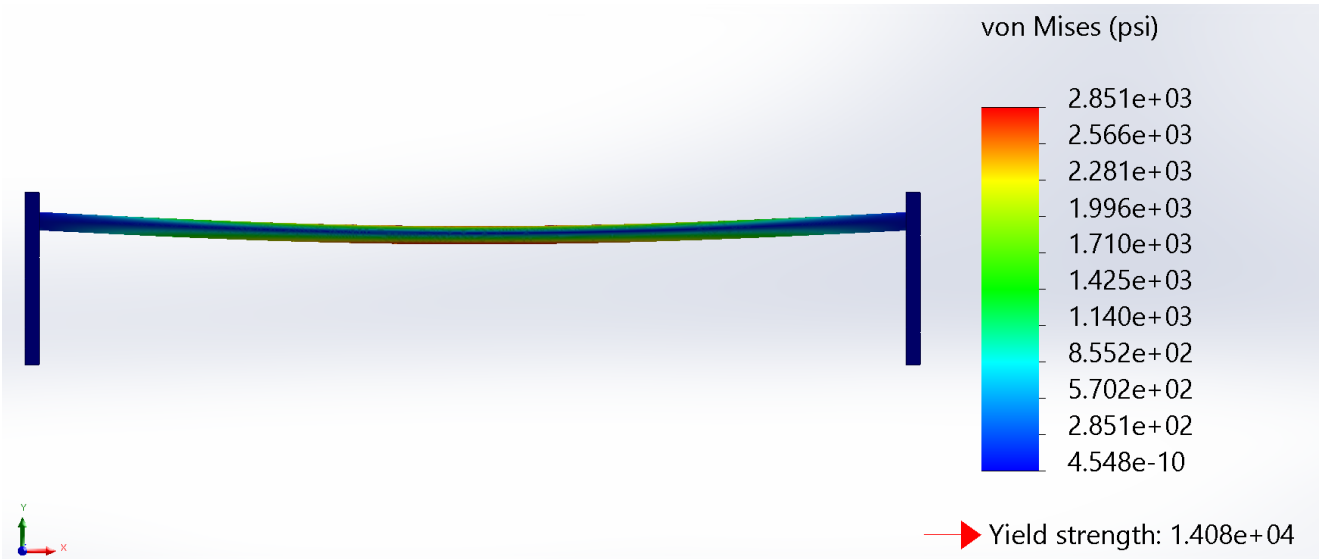
Appendix Figure C1. Supplementary axial stress plot from the SolidWorks Simulation study showing bottom side of rod.



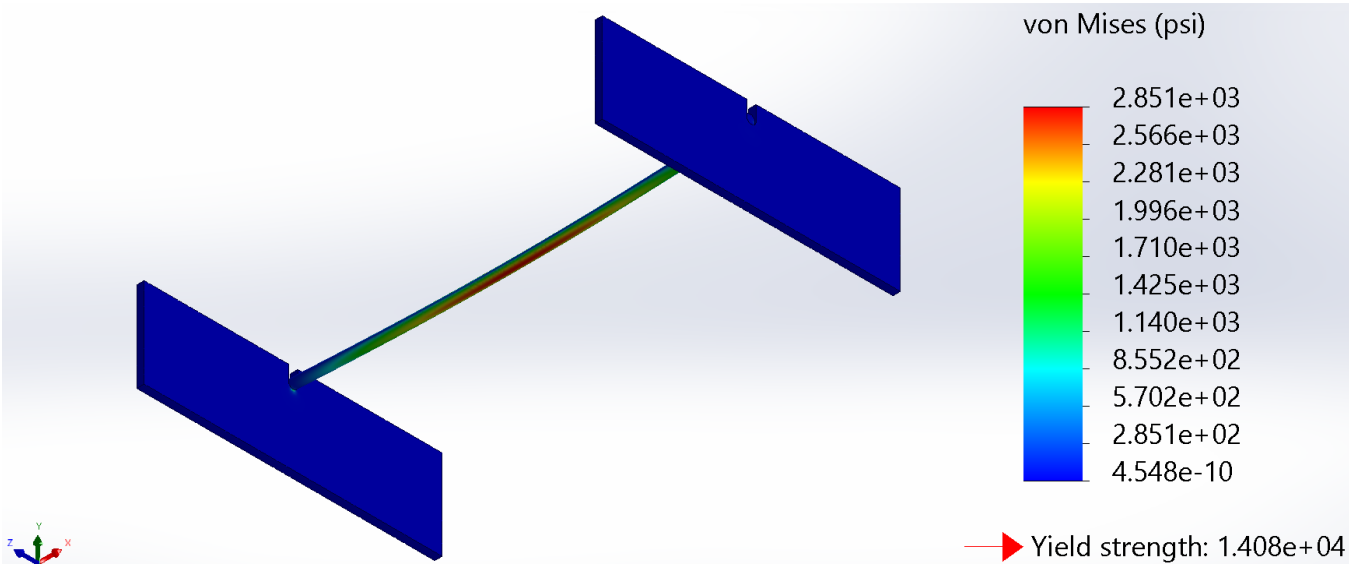
Appendix Figure C2. Model of reaction force at A from the SolidWorks Simulation study.



Appendix Figure C3. Model of reaction force at B from the SolidWorks Simulation study.



Appendix Figure C4. Von Mises stress plot from SolidWorks Simulation study, front view.



Appendix Figure C5. Von Mises stress plot from SolidWorks Simulation study, bottom view.